Project: Worked examples
Title: Voltage drop and I²R loss
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Calculates the voltage drop, the conductor power loss and the estimated conductor temperature of a DC, single-phase or three-phase circuit. The drop limit is a user input, because the limits of BS 7671 are not embedded.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Conductor resistance at its operating temperature, for n conductors in parallel | $R = \rho_T L/(A\,n)$, with $\rho_T = \rho_{20}\,[1 + \alpha\,(T_c - 20)]$ |
| Resistance and reactance from per-length figures instead | $R = R' L/n$; $X = X' L/n$ |
| Voltage drop on a DC circuit | $\Delta V = 2\,I\,R$ |
| Voltage drop on a single-phase AC circuit | $\Delta V = 2\,I\,(R\cos\varphi + X\sin\varphi)$ |
| Voltage drop on a three-phase AC circuit | $\Delta V = \sqrt{3}\,I\,(R\cos\varphi + X\sin\varphi)$ |
| Voltage drop from a tabulated mV/A/m figure, which already carries the 2 or √3 | $\Delta V = (\text{mV/A/m})\,I\,L / 1000$ |
| Power lost in the cable | $P_{loss} = 2\,I^2\,R$ (DC and single-phase); $3\,I^2\,R$ (three-phase) |
| Estimated conductor temperature at this current | $T_{c,est} = T_{amb} + (T_{rated} - T_{amb})(I/I_z)^2$ |

Basis: Voltage drop in consumers' installations[1](reg. 525); Voltage drop (mV/A/m tables and the temperature-correction method)[1](App. 4 §6); Resistance, resistivity and temperature coefficient[2](Ch. on DC and AC circuits, 12th ed. (edition to be checked)).

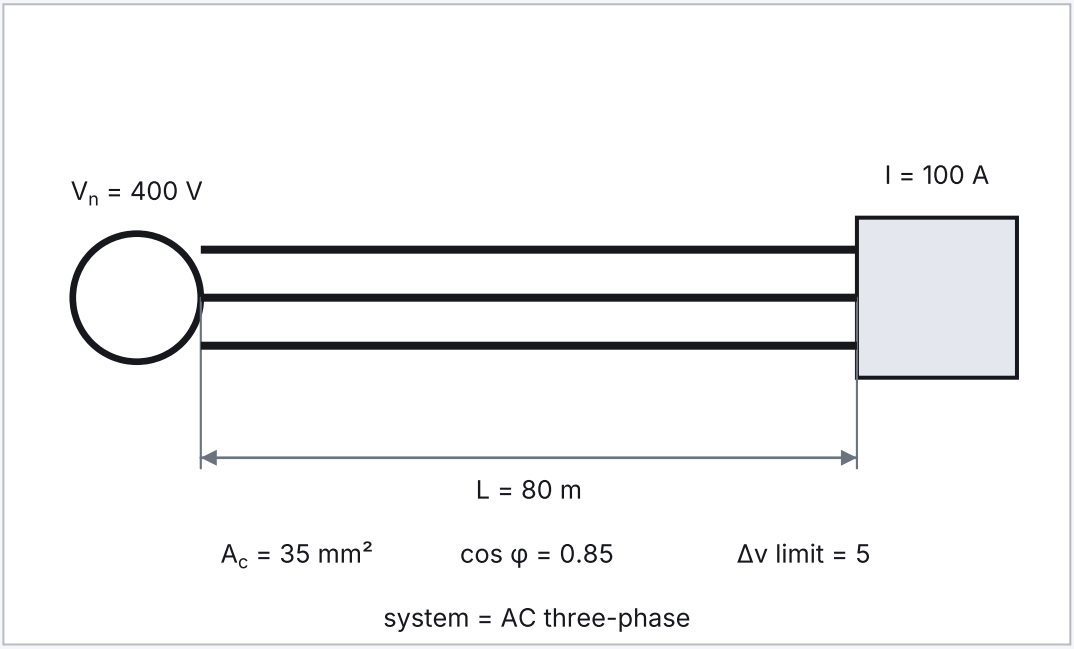



Figure 1 — A three-phase circuit: the supply, three line conductors of route length L, and the load current I. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| system | System | AC three-phase | | assumption: Design convention. | any |
| data | Conductor data from | resistivity and area | | assumption: Design convention. | any |
| L | Route length (one way) | 80 | m | drawing: DRG-E-120 rev B | 0.1 … 5000 m |
| $n_{par}$ | Parallel conductors per phase | 1 | | assumption: Design convention. | 1 … 6 |
| $A_c$ | Conductor cross-section | 35 | mm² | spec: Cable schedule CS-01: 35 mm² Cu XLPE/SWA | 1 … 1000 mm² |
| α | Temperature coefficient of resistance | 0.00393 | | reference: Reference record: annealed copper 0.01724 Ω·mm²/m, 0.00393/K (Hughes) | 0.003 … 0.0045 |
| $X_{km}$ | Reactance per km per conductor | 0.08 | Ω/km | spec: Cable datasheet: 0.08 Ω/km | 0 … 1 Ω/km |
| $T_{rated}$ | Conductor rated temperature | 70 | °C | assumption: Design convention. | 60 … 90 °C |
| $\rho_{20}$ | Resistivity at 20 °C, the effective value for the conductor resistance (default 0.0194 Ω·mm²/m: the largest class 2 maximum conductor resistance of the cable standard for 1.5 to 120 mm², so the drop is not below the cable tables; pure annealed copper is 0.01724, aluminium 0.0282 Ω·mm²/m) | 0.01724 | Ω·mm²/m | reference: Reference record: annealed copper 0.01724 Ω·mm²/m, 0.00393/K (Hughes) | 0.01 … 0.05 Ω·mm²/m |
| $V_n$ | Nominal voltage (line-to-line for three-phase) | 400 | V | assumption: Design convention. | 12 … 1000 V |
| I | Design current | 100 | A | spec: Motor schedule MS-01: 100 A full-load current | 0.1 … 5000 A |
| $cos_{phi}$ | Power factor (AC) | 0.85 | | assumption: Design convention. | 0.5 … 1 |
| $T_c$ | Conductor operating temperature for the resistance | 70 | °C | assumption: Design convention. | 20 … 90 °C |
| $T_{amb}$ | Ambient temperature | 30 | °C | assumption: Design convention. | 0 … 60 °C |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| $I_z$ | Tabulated current-carrying capacity for the installed conditions (temperature estimate) | 119 | A | spec: Cable schedule CS-01: I_z = 119 A for the installation method | 1 … 5000 A |
| $dv_{limit}$ | Voltage-drop limit | 5 | | spec: Project spec ES-3 §4: 5 % (BS 7671 Appendix 4 §6.4 default for other uses) | 1 … 10 |

## 3. Assumptions

A1. The load is balanced across the phases. (Standard engineering approximation.)
A2. The conductor reactance is the entered X′ per kilometre times the length. (Standard engineering approximation.)
A3. The voltage drop is the arithmetic sum R cos φ + X sin φ, not the vector difference. (Standard engineering approximation.)
A4. A mV/A/m figure is used as tabulated, with its own temperature and system convention. (Cable table convention.)
A5. The estimated conductor temperature scales the rated temperature rise with $(I/I_z)^2$. (BS 7671:2018 Appendix 4 approach, formula only.)
A6. $I_z$ is the tabulated capacity for the installation method, with its correction factors applied. (BS 7671:2018 Appendix 4 approach, formula only.)
A7. When the estimate is above $T_c$, the resistance and the drop are computed at the estimate. (Keeps the drop on the safe side.)
A8. system (System) is taken as AC three-phase. (Design convention.)
A9. data (Conductor data from) is taken as resistivity and area. (Design convention.)
A10. $n_{par}$ (Parallel conductors per phase) is taken as 1. (Design convention.)
A11. $T_{rated}$ (Conductor rated temperature) is taken as 70 °C. (Design convention.)
A12. $V_n$ (Nominal voltage (line-to-line for three-phase)) is taken as 400 V. (Design convention.)
A13. $cos_{phi}$ (Power factor (AC)) is taken as 0.85. (Design convention.)
A14. $T_c$ (Conductor operating temperature for the resistance) is taken as 70 °C. (Design convention.)
A15. $T_{amb}$ (Ambient temperature) is taken as 30 °C. (Design convention.)

## 4. Working

[9] $sin_{phi} = sqrt(1 - cos_{phi}^2)$
= sqrt(1 - 0.85^2)
= 0.52678
sin φ

[18] $u_{Iz} = I / I_z$
= 100 A / 119 A
= 0.84034
Load against the current-carrying capacity

[19] $T_{c,est} = T_{amb} + (T_{rated} - T_{amb}) * u_{Iz}^2$ [1](App. 4 §6 form)
= 30 degC + (70 degC − 30 degC) × 0.84034^2
= 58.247 degC
Estimated conductor temperature

[20] $\rho_T = \rho_{20} * (1 + \alpha * (T_c - 20\ degC) / (1\ K))$
= 0.01724 ohm·mm²/m × (1 + 0.00393 × (70 degC − 20 degC) / 1 K)
= 0.020628 ohm·mm²/m
Resistivity at the operating temperature

[21] $R = \rho_T * L / (A_c * n_{par})$
= 0.020628 ohm·mm²/m × 80 m / (35 mm² × 1)
= 0.047149 ohm
Conductor resistance (one conductor, parallel paths combined)
[22] $X = X_{km} * L / n_{par}$
= 0.08 ohm/km × 80 m / 1
= 0.0064 ohm
Conductor reactance
[23] $dV = sqrt(3) * I * (R * cos_{phi} + X * sin_{phi})$
= sqrt(3) × 100 A × (0.047149 ohm × 0.85 + 0.0064 ohm × 0.52678)
= 7.5254 V
Voltage drop
[24] $dV_{pct} = 100 * dV / V_n$
= 100 × 7.5254 V / 400 V
= 1.8814
Voltage drop, per cent
[25] $P_{loss} = 3 * I^2 * R$
= 3 × (100 A)^2 × 0.047149 ohm
= 1.4145 kW
Conductor power loss
[26] $u_{dv} = dV_{pct} / dv_{limit}$
= 1.8814 / 5
= 0.37627
Drop utilisation against the limit
[27] $dV_{max} = dv_{limit} / 100 * V_n$
= 5 / 100 × 400 V
= 20 V
Largest voltage drop the limit allows

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| dV | 7.525 V | Voltage drop |
| $dV_{pct}$ | 1.8814 | Voltage drop (%) |
| $P_{loss}$ | 1.414 kW | Power loss |
| $T_{c,est}$ | 58.25 °C | Estimated conductor temperature |
| R | 0.04715 Ω | Resistance |
| X | 0.006400 Ω | Reactance |
| $u_{dv}$ | 0.38 | Drop utilisation against the limit |
| $u_{Iz}$ | 0.84 | Load against the current-carrying capacity (governing) |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Design current against the current-carrying capacity | 100 A | 119 A | 0.84 | [1](7671:2018 reg. 523) |
| PASS | Voltage drop against the limit | 7.525 V | 20 V | 0.38 | [1](7671:2018 reg. 525) |

## 7. Applicability

None.

## 8. Verification

```
Template elec.voltage-drop@1.5.0 Content hash a149837c72d9fb56 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. Hughes E. Hughes Electrical and Electronic Technology. 12th ed. Pearson; 2016.